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3CS-A

Part 1: Familiarizing the Blender Interface
Step 1: Launch Blender
1. Open Blender.
2. Select General from the startup screen.
3. Delete the default cube:
o Select the cube.
o Press X → Delete.

Part 2: Creating the Table
Step 2: Create the Tabletop
1. Press Shift + A.
2. Select Mesh → Cube.
3. Rename the object Tabletop.
Step 3: Resize the Tabletop
1. Press S (Scale).
2. Scale along the X and Y axis to make it wider.
3. Scale along the Z axis to make it thinner.
Example dimensions (optional):
• X: 2.5
• Y: 1.5
• Z: 0.1

Step 4: Create the Table Legs
1. Press Shift + A → Mesh → Cube.
2. Rename the object Table Leg.
Step 5: Resize the Leg
1. Press S.
2. Scale:
o X: 0.1
o Y: 0.1
o Z: 1
Step 6: Position the Leg
1. Press G (Move).
2. Place the leg at one corner under the tabletop.
Step 7: Duplicate the Legs
1. Select the leg.
2. Press Shift + D to duplicate.
3. Move the duplicated legs to the remaining three corners.
Now you should have 4 table legs.
Part 3: Creating the Chair
Step 8: Create the Seat

1. Press Shift + A → Mesh → Cube.
2. Rename it Chair Seat.

Step 9: Resize the Seat

Scale the cube:

- X: 0.8
- Y: 0.8
- Z: 0.1

Move it slightly above the floor.

Step 10: Create Chair Legs

1. Add a cube (Shift + A → Cube).
2. Scale it to form a leg:
   - X: 0.07
   - Y: 0.07
   - Z: 0.8
3. Move it to one corner of the seat.
4. Duplicate (Shift + D) until there are 4 chair legs.

Step 11: Create the Backrest

1. Press Shift + A → Mesh → Cube.
2. Scale it to form the backrest:
   - X: 0.8
   - Y: 0.1
   - Z: 1
3. Move it behind the seat.

Part 4: Applying Materials

Step 12: Add Basic Materials

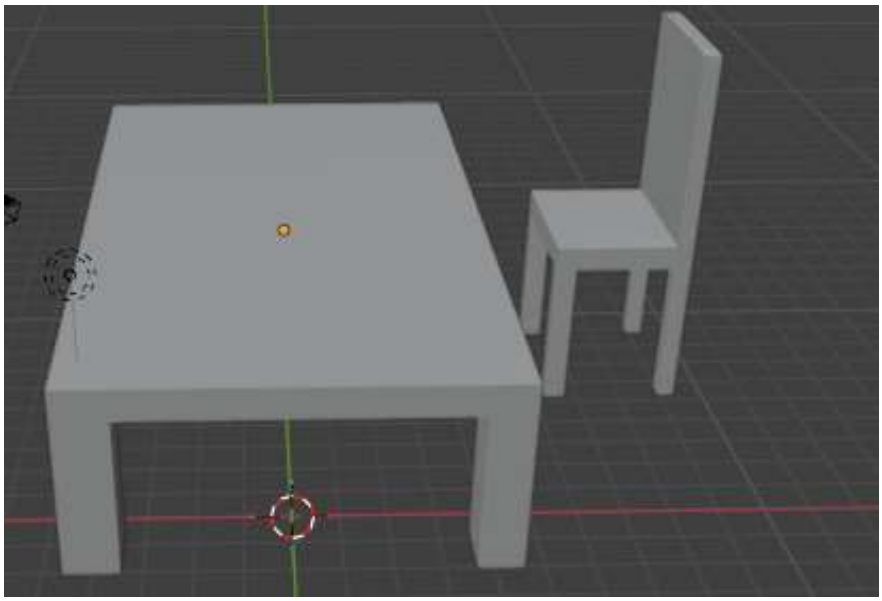
1. Select the Tabletop.
2. Go to the Material Properties panel.
3. Click New.
4. Choose a wood-like color.

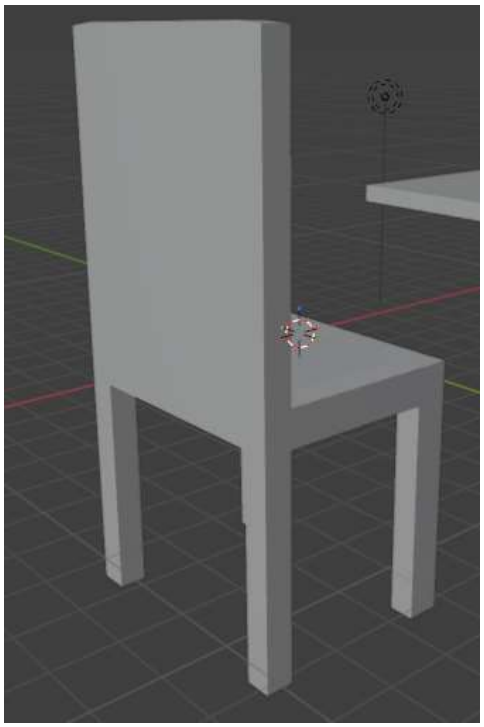
Repeat for the chair.

Part 5: Viewing the Final Model

Step 13: Render the Scene

1. Press Z.
2. Choose Rendered View.
3. Add light if necessary:
   - Shift + A → Light → Point Light.





Guide Questions:

1. What primitive object was used to create the table and chair components?
When creating a table and chair in 3D modeling, designers usually begin with simple shapes called primitive objects. These are the building blocks of any 3D model, and the most commonly used one is the cube. A cube is very flexible because it can be stretched, scaled, and reshaped into different parts of furniture. For example, the tabletop often starts as a flattened cube, which is then adjusted to the desired thickness and size. The legs of the table can also come from cubes that are stretched downward to form long supports. In some cases, cylinders are used instead, especially if the design calls for rounded legs or a smoother look. For chairs, the seat is usually made from a cube that is scaled to be wide and thin. The backrest can also be another cube that is rotated and positioned vertically. Designers often combine multiple cubes to form more complex shapes, like armrests or supports. Using these primitives makes the modeling process easier and more organized. It allows the creator to focus on shaping rather than starting from scratch. These simple objects can be modified through tools like scaling, rotating, and extruding. Over time, what started as basic shapes becomes a recognizable piece of furniture. This method is efficient and widely used in beginner and professional workflows alike. It also helps maintain symmetry and proper proportions in the model. By starting with primitives, designers can experiment with structure and form more freely. In the end, even complex models are often just combinations of these simple starting shapes.

2. Which transformation tool is used to resize objects?

The transformation tool used to resize objects in 3D modeling is called the Scale tool. This tool allows you to make an object bigger or smaller depending on your needs. When working on a model, resizing is something you do frequently to adjust proportions. The Scale tool can change the size of an object uniformly, meaning all sides grow or shrink at the same rate. It can also scale along specific axes, like making something taller without making it wider. This gives you more control over the shape of your object. For example, when modeling a chair leg, you might scale it vertically to make it longer. The Scale tool is usually easy to access, often through a toolbar or a shortcut key. It works by dragging handles or inputting values to define the new size. This tool is essential because models rarely come out the perfect size on the first try. Designers constantly adjust objects to fit their vision. Scaling helps maintain balance and proportion in the overall scene. Without it, everything would look either too large or too small compared to other elements. It also helps in aligning objects with real-world measurements when needed. The Scale tool is simple but very powerful. It is one of the first tools beginners learn when starting 3D modeling. As you gain experience, you realize how often you rely on it. In almost every project, resizing objects is part of the process. Because of this, the Scale tool becomes one of the most important tools in your workflow.

3. What shortcut key is used to duplicate an object?

In many 3D modeling programs, the shortcut key used to duplicate an object is Shift + D. This shortcut is especially common in Blender, which is widely used by beginners and professionals. Duplicating an object means creating a copy of it without having to build it again from scratch. This is very helpful when working on objects that repeat, like chair legs or table supports. Instead of modeling each leg individually, you can create one and duplicate it several times. This saves time and keeps the design consistent. When you press Shift + D, a copy of the selected object appears and can be moved into position. You can then place it exactly where you need it. This process is quick and efficient, making your workflow smoother. It also reduces errors because all duplicated parts start with the same shape and size. In furniture modeling, duplication is used constantly. For example, a table usually has four legs, and duplicating ensures they are identical. The same applies to chairs, railings, or any repeating structures. Learning this shortcut helps speed up your work significantly. It allows you to focus more on creativity rather than repetitive tasks. Many designers rely on duplication as a core part of their process. Over time, it becomes second nature to use this command. It also works well with other tools like scaling and rotating. By combining duplication with transformations, you can build complex models quickly. Overall, this shortcut is simple but extremely useful in 3D modeling.

4. Why is scaling important in 3D modeling?
Scaling is important in 3D modeling because it helps ensure that objects look realistic and properly proportioned. When creating models, it is not enough to simply shape them correctly—they also need to be the right size. If something is too big or too small, it can make the entire scene feel unrealistic. Scaling allows you to adjust objects so they fit well with other elements. For example, a chair should not be taller than a table in a normal setup. By using scaling, you can correct these kinds of issues easily. It also helps when working with measurements or references. Designers often base their models on real-world dimensions, and scaling helps match those accurately. Another reason scaling is important is that it allows flexibility during the design process. You may change your mind about the size of an object, and scaling makes it easy to adjust without starting over. It also helps create visual balance in a scene. Objects that are properly scaled look more natural and pleasing to the eye. In animation and rendering, correct scale is even more important. Lighting, shadows, and physics can behave differently depending on the size of objects. Scaling ensures that everything interacts correctly. It also helps maintain consistency across a project. Without scaling, models would look disconnected and unrealistic. It is a basic tool, but it plays a major role in the final outcome. Every designer uses scaling throughout the entire modeling process. It is one of the key factors in creating believable and professional-looking 3D work.

5. How does applying materials improve the appearance of a 3D model?
Applying materials greatly improves the appearance of a 3D model by making it look more realistic and visually interesting. Without materials, a model usually appears as a plain gray shape, which does not reflect how objects look in real life. Materials add color, texture, and surface properties to the model. For example, a wooden table can be given a wood texture to make it look natural. A metal chair can have a shiny or reflective surface to simulate real metal. These details make a big difference in how the model is perceived. Materials also interact with lighting, which enhances realism. Light can reflect, absorb, or scatter differently depending on the material applied. This creates highlights, shadows, and depth in the scene. Applying materials helps bring life to your model. It allows viewers to better understand what the object is made of. It also adds personality and style to the design. For example, you can create a modern look with glossy finishes or a rustic feel with rough textures. Materials are essential for rendering, where the final image is produced. Without them, the render would look flat and unfinished. They also help communicate the purpose and quality of the object. In professional work, materials are just as important as the shape of the model. They complete the overall look and feel of the design. Learning how to apply and adjust materials is a key skill in 3D modeling. It transforms simple shapes into believable objects that resemble real-world items.